

A New BMS Architecture Based on Cell Redundancy

Antonio Manenti, Andrea Abba, Alessandro Merati, Sergio M. Savaresi, *Member, IEEE*, and Angelo Geraci, *Senior Member, IEEE*

Abstract—Due to their great energy density, high nominal voltage, and lack of memory effect, Li-ion cells play a more crucial role in electric vehicle (EV) applications. Unfortunately, if not correctly handled, this chemistry can quickly bring loss in performance and produce potentially dangerous situations. Battery management systems (BMSs) are a practical way to manage these battery packs and improve their efficiency. In this paper, we propose an original BMS architecture particularly suited for light-EV applications based on the concept of redundant cell, i.e., a technique that dynamically disconnects a cell in the battery pack for optimal balancing purposes. The proposed BMS is very flexible, economical, stable, and space saving.

Index Terms—Battery management system (BMS), cell-voltage-balancing method, light electric vehicles (EVs), Li-ion battery pack.

I. INTRODUCTION

THE priority of electric vehicles (EVs) as a primary factor of sustainable mobility is widely recognized over the world also in the low-power two-wheel segment, i.e., scooters and bikes.

Lithium-ion (Li-ion) batteries, due to their great energy density, high terminal voltage, and lack of memory effect, have been commonly used for several years in portable applications. Despite the higher terminal voltage provided by this chemistry (if compared with others), a string constituted by more than one cell has to be used to supply high-power demanding applications that require voltage from several tens up to hundreds of volts. In this configuration, different internal resistance, degradation, and temperature gradient along the string pack can cause a capacity imbalance in the form of voltage variations, determining overcharge of some cells or undercharge of some others. As other chemistries, the voltage of Li-ion batteries has to be maintained within specific minimum and maximum limits depending on the specific cell. The problem is that they do not have a natural gassing mechanism, so exceeding these limits could entail severe safety problems with consequent explosion risk [1]. Moreover, the repetition of charge/discharge cycles increases cell voltage imbalance, which is particularly detrimental in EV applications because of charge/discharge dynamics. In fact, unlike standard applications where almost

complete charge/discharge phases are followed one after another, in EV use (except for a plug-in hybrid EV), charge/discharge phases are partial and interleaved. This is due to the fact that discharge occurs during acceleration and the battery is able to recharge when braking [2]. For all the presented reasons, a charge/discharge equalization procedure that is able to improve the maintenance in time of cell efficiency has to be adopted.

The easiest solution should be to disconnect the load when a single cell reaches its maximum/minimum value. Unfortunately, this determines a strong reduction of the pack capacity. Many other equalization solutions have been proposed in order to prevent cell imbalance during the charge/discharge of series-connected battery packs [3]–[7]. Each of these solutions aims to a specific goal such as cost reduction, space saving [8], robustness, and so on.

In agreement with classification proposed in [9], we focused on nondissipative bidirectional equalizers. First of all, this means that the proposed battery management system (BMS) does not use extra power to prevent imbalance, thereby increasing the overall efficiency. In addition, it continues to work during both charge and discharge operations, improving balancing effectiveness. We present a new BMS architecture that is suited for linear packs made up of a maximum of ten Li-ion cells for low–medium power applications (in the range from 100 to 500 W). This solution can be oriented to light EVs (i.e., electric bicycles), which usually need low–medium voltage (24–36 V) and current (10 A) associated to a simple, inexpensive, and space-saving layout. In fact, no extra components such as dc–dc converters, power inductors, and transformers are needed by the BMS. The proposed BMS strategy is based on the availability of an additional cell that can be continuously disconnected from the load. In this way, the bypassed cell is isolated to be optimally balanced while the remainder of the pack continues to provide energy to the load. At any time, while the pack works, at least one cell is disconnected from the string.

There are three main features of the proposed architecture. First, balancing is active in both the charge and discharge phases. Second, the BMS layout is considerably space saving, which is a primary task for two-wheel vehicle applications that are often subjected to hard space constraints. Third, the solution offers intrinsic hardness to a cell fault. In fact, if the pack should contain a cell with a severe fault, the BMS can identify and bypass it permanently.

II. ARCHITECTURE

As previously indicated, in the literature, many bidirectional nondissipative balancing architectures have been presented

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The authors are with the Dipartimento di Elettronica e Informazione, Politecnico di Milano, 20133 Milano, Italy (e-mail: manenti@elet.polimi.it; abba@elet.polimi.it; alessandro.merati@mail.polimi.it; savaresi@elet.polimi.it; geraci@elet.polimi.it).

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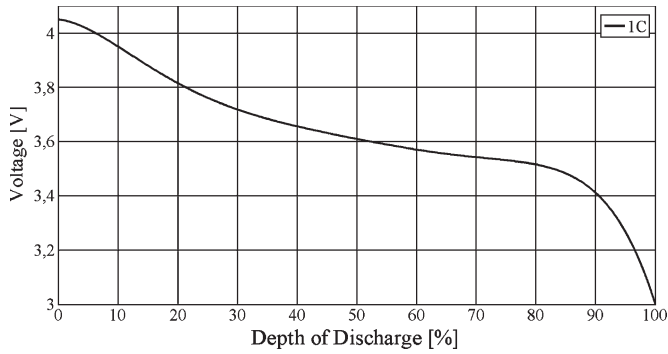


Fig. 1. Typical discharge profile of a lithium-cobalt cell at 1 C.

[10]. Most of them use different types of dc-dc converters in Čuk [11] or buck-boost [12] topologies. The drawback of these solutions is that they require fast, expensive, and bulky power switches, inductors, and diodes. Furthermore, even if those architectures could seem plain, the control units are usually extremely complex since they have to manage many high-frequency and well-timed pulsewidth modulation (PWM) control signals.

The idea to implement a system on which cells of the string can be independently disconnected by means of switches is not new in the literature, but only in the presented work, it is used along with a state-of-charge (SoC) balancing algorithm and a redundant cell. In [13], in fact, the switch structure was mainly used to prevent potentially dangerous situations and to bypass damaged cells. Moreover, the presented architecture is able to handle a higher power load, performing a continuous cell-balancing action using more versatile and faster MOSFET-based switches.

The BMS has been designed bearing in mind a general-purpose application, both in terms of cell type and operating conditions. In particular, the realized prototype is made for seven standard lithium-ion cells with the following features:

- 1) 4.2 V of full charge voltage;
- 2) 4400 mAh of capacity;
- 3) 10 A of maximum continuous current load;
- 4) 3 A of maximum charge current.

This means that the pack can supply a nominal voltage of about 24 V (six cells connected at the same time). Fig. 1 shows a typical discharge profile of the used cell.

The block diagram of the proposed BMS architecture is shown in Fig. 2. The architecture is constituted of three main stages: the switch network, the data acquisition block, and the control unit.

Ancillary features have also been implemented to make the system more versatile. For instance, a load presence controller and a low-power buck converter have been added. In particular, the latter permits the pack to manage its own charge, applying an inexpensive unlimited voltage generator on two external terminals. In the following, the three main blocks of the BMS will be analyzed.

A. Switch Network

The switch network is the fundamental section of the system since it is critical both for safety and efficiency issues. Even if,

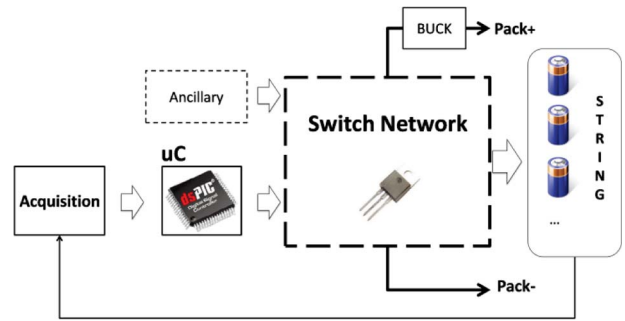


Fig. 2. Block diagram of the proposed BMS architecture. The cores of the system are the data acquisition block ("Acquisition"), the control unit (the microcontroller "uC"), and the "Switch Network."

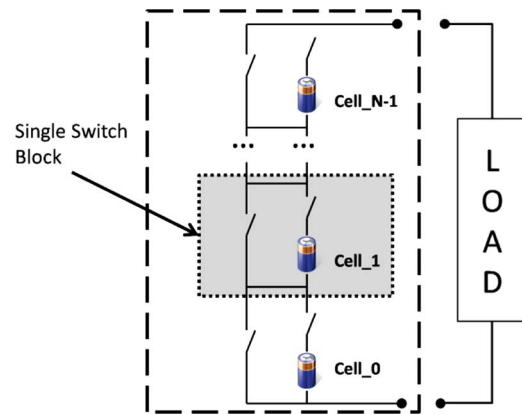


Fig. 3. Basic representation of the switch network.

at a first glance, the structure and the operation modality of the circuit are simple (Fig. 3), the implementation revealed several tricky issues both in the layout and the management algorithm. In details, they are the following.

- 1) Since switches are connected in series on the load current path, their resistance directly impacts on the performances (in terms of efficiency) of the system.
- 2) Switches have to be able to interrupt current flow in both directions. In fact, the system must perform cell balancing both in charge phase, i.e., current flowing from the external terminal to the cells, and in discharge phase, i.e., current flowing from the cells to the external load.
- 3) Topic of the implementation is to minimize the number of components, which also prevents the necessity to use specific drivers of switches.
- 4) The system has to be safe in every operative condition, e.g., even if a single cell be physically taken out from the pack or the microcontroller stops for a while.

All these constraints make the design of the stage critical both from the layout and component choice points of view.

A simplified final schematic of the realized single switch between two adjacent cells is shown in Fig. 4. Each cell has two switches: one serially connected to the cell and the other in parallel to the cascade cell switch. Every single switch is constituted by two MOSFETs; in particular, connection switches are constituted by n-channel MOSFETs (NMOSs), whereas bypass switches use p-channel MOSFETs (PMOSs).

In this way, during normal operation, $N - 1$ cells are connected

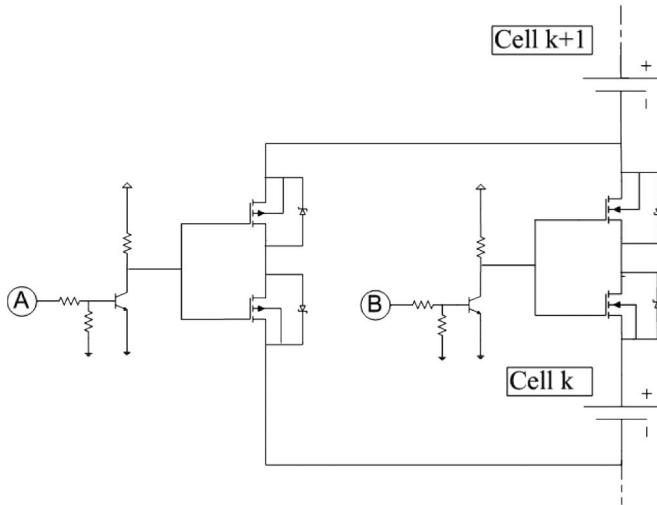


Fig. 4. Simplified final schematic of a single switch block. A and B are the gates for control of the switch status by the control unit.

TABLE I
MAIN CHARACTERISTICS OF THE MOS TRANSISTORS
USED FOR THE PROTOTYPE

	STS12NH3LL	IRF7425
Type	NMOS	PMOS
R _{ds-on} [mΩ]	10	13
I _{D max} [A]	12	15
R _{thja} [°C/W]	47	50

by NMOS switches, and only one cell is bypassed through a PMOS switch. This represents a significant advantage for the whole system efficiency since NMOSs guarantee smaller ON-state resistance than PMOSs.

To guarantee a stable and safe configuration also when command signals on gates of MOSFETs are left floating due to accidental control unit disconnection or initialization, a protection system has been added. It is based on bipolar junction transistors (BJTs) in open-collector configuration in addition with pull-up resistors. It should be noted that no specific driver circuits have been used, helping maintaining the cost low.

For border cells, i.e., bottom cell 0 and top cell $N - 1$, the switch architecture differs from the others. In fact, in these cases, using the usual structure would have required control signals overcoming rail voltage levels. Therefore, the bottom cell has been designed using both NMOS-based switches, whereas those of the top cell are all PMOS based. This solution makes the bottom cell easier to control and more efficient due to the switch's greater conductivity. On the contrary, the top cell is less efficient and requires further attention. In fact, during system initialization, the terminals remain floating, opening both the switches. The problem has been solved by adding a Schottky diode at the top of the pack, which provides automatic connection of cell $N - 2$ to the output terminal in those cases. Table I reports the main features of these MOSFETs used for the prototype.

Of course, during pack reconfiguration, the terminal voltage experiences some jumps. By the way, this is not an issue for at least two reasons. First, these jumps are related to

different dynamics if compared with those of the load (i.e., electric motor). In fact, the entire reconfiguration process just lasts several hundreds of microseconds, whereas standard loads have much slower dynamics (tenth of milliseconds). Therefore, basically, the load is not sensitive to these phenomena. The second reason is related to the fact that, everywhere, in a well-designed system, between the BMS and the load, a load control system is placed. This is usually adopted to match the battery voltage (that can strongly vary with pack SoC) with the load requirements (that depend on the operative conditions). These systems usually have a bandwidth that is high enough to handle and level voltage jumps occurring during pack reconfiguration.

B. Acquisition

The BMS needs to acquire voltage on each cell, overall pack voltage, and charge/discharge current. In order to minimize circuit complexity, a microcontroller internal analog-to-digital converter (ADC) has been used as the acquisition gate. Despite the clear advantages concerning space and cost saving that this solution provides, it can create some drawbacks since standard internal ADCs have only 10- or 12-b resolution. The worst case in acquisition operation occurs when considering the top cell (i.e., the pack voltage), where the voltage to be sampled can approach 25 V if all connected cells are fully charged. In this case, considering only quantization error and a 12-b ADC, the theoretical resolution that can be achieved is about 6 mV. This is on the order of the value generally specified as a minimum resolution in cell specifications for maximum charge voltage control that is able to guarantee good capacity and preserve safety. The necessary resolution can be achieved in two ways.

A "hardware" solution consists of sampling a properly conditioned signal from cells. This solution is based on the consideration that cell voltage does not span between 0 and 5 V (standard ADC input range) but actually between 2.5 and 4.2 V. Reporting this evidence to all cells, the voltage on the top cell varies from 15 to 25 V, turning to a real full-scale range (FSR) of only 10 V. This reduces the FSR by 2.5, increasing the resolution to the same factor. A value of 2.4 mV is compliant with the specification of standard cells.

A "software" solution is also available. By oversampling the signal, it is possible to reduce quantization noise. Since a significant time interval comes between consecutive acquisitions (order of seconds), the microcontroller can acquire more times the same signal and extract its mean value. It is well known that the M -times repetition of the signal acquisition improves the signal-to-noise ratio of the consequent measurement of the factor $\sqrt{M}$.

Both oversampling by a factor of 16 and considering the hardware solution enhancement, it is possible to achieve a 600- μ V resolution. That is more than one order of magnitude less with respect to the value specified as minimum for standard cells. These considerations made for a seven-cell configuration become even more valid when considering a ten-cell string.

The voltage on each cell is acquired differentially by means of an instrumentation amplifier (INA), making the measurement independent from the cell connection status. In order to preserve the number of microcontroller inputs used and the

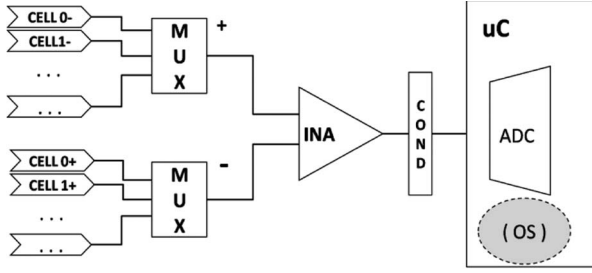


Fig. 5. Acquisition stage block diagram.

number of used INAs, two eight-channel multiplexers have been used. While one multiplexer sends data from positive terminals to the INA, the other sends the negative. The block diagram of the acquisition stage is shown in Fig. 5.

The current measurement is obtained by using the LTC6103 integrated circuit (IC) from Linear Technology. Basically, it is a low-cost current-sense amplifier that permits measuring both positive and negative current values with no need of negative power supply.

C. Microcontroller

Microcontroller is the “smart core” of the entire system. It performs signal acquisition by means of its internal ADC, manages all switches, processes sampled signals, and, in cases, evaluates the SoC of each cell according to the chosen method.

Microcontroller choice has been driven by the following three main factors:

- 1) need large pinout to control switches, as each cell requires two I/Os to drive its switches;
- 2) high-resolution (at least 12 b) internal ADC; no strict timing constraint required;
- 3) availability of necessary signal processing resources.

The chosen device is a dsPIC microprocessor of Microchip Devices (dsPIC30F3014) that is specifically designed for high processing throughput and very low power consumption which limit the device’s cost. Due to its very low power consumption, the microprocessor can be powered by the battery pack through a low-dropout voltage regulator without weight on the BMS efficiency. In fact, the absorbed mean current (less than 1 mA) is negligible if compared with the amount required from the load.

III. BALANCING ALGORITHM

The algorithm that manages cell connection or bypass represents a crucial part of the system. It directly impacts performances in terms of balance effectiveness and convergence speed.

The algorithm has been developed bearing in mind the achievement of the best performance with the simplest algorithm architecture. The flowchart of the developed algorithm is shown in Fig. 6.

The first algorithm step is data acquisition, where all needed signals are sampled (cell voltage, pack voltage, and current). Then, if used, the oversampling procedure is performed. The core of the algorithm is the management of the cell to be

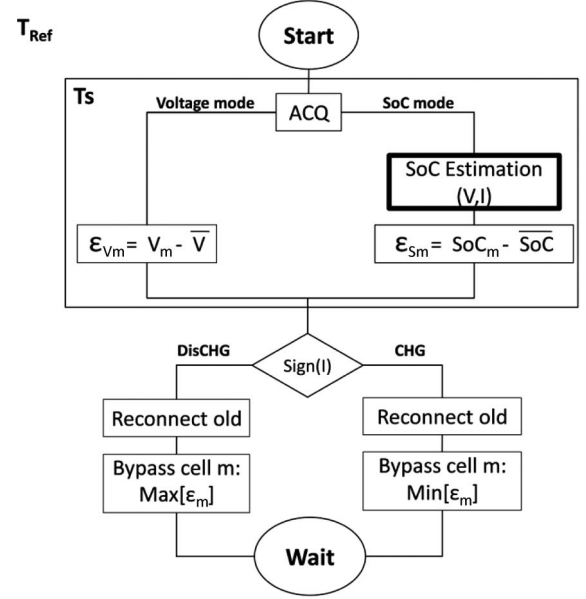


Fig. 6. Developed algorithm flowchart.

disconnected. Two decision strategies have been explored: a voltage analysis and a SoC analysis.

While voltage analysis simply consists of monitoring the voltage across each cell with no need for additional acquisitions and processing, SoC analysis is more complex as it requires both voltage and current measurements to estimate the SoC of each cell. Several techniques for SoC estimation can be found in the literature [14]–[17]. Unfortunately, most of them (e.g., open-circuit voltage measurement, impedance spectroscopy, neural networks, and fuzzy logic) are not suited to be used in this application. In fact, some of them require battery disconnection or long resting times, whereas some others need a high amount of memory and computation resources (often not available in embedded systems). In this case, the “mixed estimation algorithm” presented in [18] (and recalled in the next section) has been chosen. In any case, whatever the adopted decision strategy is, the algorithm calculates the mean value of the control variable over all the cells, and for each m th cell, it calculates the deviation ϵ_m of its control variable value with respect to the mean one. In voltage-mode operation, this deviation is

$$\epsilon_{Vm} = V_m - \bar{V} \quad (1)$$

while in SoC-mode operation, the deviation looks like

$$\epsilon_{Sm} = \text{SoC}_m - \bar{\text{SoC}}. \quad (2)$$

Checking the current sign, the algorithm establishes the working phase. Then, if a discharge phase is revealed, the cell with the lowest deviation is selected for disconnection; otherwise, that with the highest deviation value is chosen. The selected cell is bypassed, and the previously bypassed one is reconnected into the string.

The algorithm runs cyclically. In particular, the first step and SoC estimation are performed with time interval T_s , whereas pack configuration is updated with time interval T_{ref} (refresh time). For proper work, $T_{\text{ref}} \geq T_s$. Lowering the T_s value

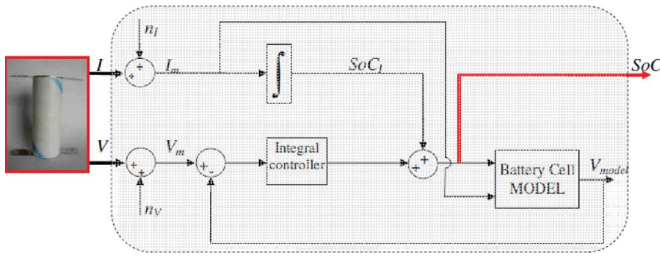


Fig. 7. Mixed SoC estimation algorithm.

increases the control bandwidth, preventing cell damage caused by overvoltages and undervoltages. On the contrary, T_{ref} is a critical parameter that hardly affects the performance of the system and whose optimal value will be discussed in the next section.

IV. SoC ESTIMATION ALGORITHM

The algorithm presented in [18] and recalled here is the synthesis of the Coulomb-counting and model-based techniques. We mix them so to keep their good characteristics and to avoid the disadvantages. Briefly, the model-based method is used to dynamically correct the estimation performed by Coulomb counting.

The Coulomb-counting method is the easiest method to estimate the SoC of a cell. Since that estimation is based on direct integration of battery current, its computation is very easy and directly connected with the definition of SoC. However, it presents some drawbacks. First of all, the initial value of SoC has to be known “*a priori*.” Second, the whole estimation is based only on the current measurement, so every flaw or inaccuracy (always present in vehicle applications) directly impacts on the final estimation. For these reasons, this technique requires a periodic reset of the algorithm.

On the other hand, a model-based method is more complicated since it requires a good cell model and the measurement of both current and voltage. Moreover, noisy acquisitions strongly limit the effectiveness of this solution.

The main idea is then to use the Coulomb-counting estimation for the basic SoC estimation; this estimation is corrected with a closed-loop control system, which tries to regulate the direct-model output voltage at the value of the actual measured voltage. The Coulomb-counting part of the SoC estimation is a sort of feed-forward component of the control variable of the control system. The adopted estimation algorithm scheme is shown in Fig. 7.

This method is well suited to be used in vehicle applications since it is able to deal with noisy measures and it does not require precise initialization. Furthermore, it is comparatively simple and can be easily implemented on embedded onboard electronics. Its effectiveness and sensitivity against model and measurement errors have been studied and reported in [19] and [20], respectively.

V. PROTOTYPING

A picture of the realized prototype is shown in Fig. 8.

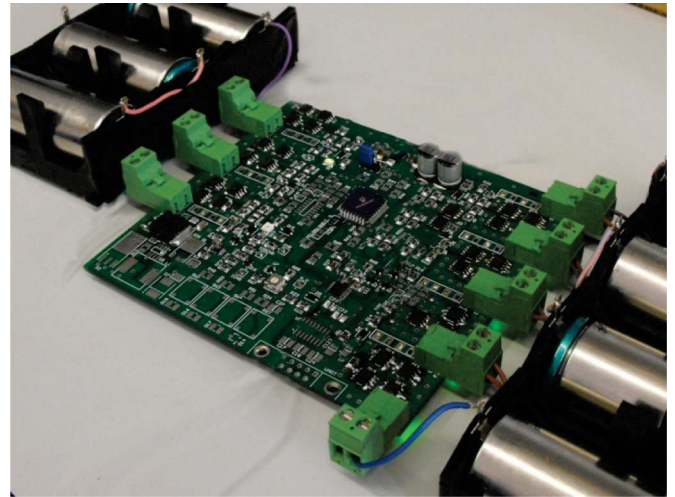


Fig. 8. Picture of the circuit implementing the proposed BMS architecture.

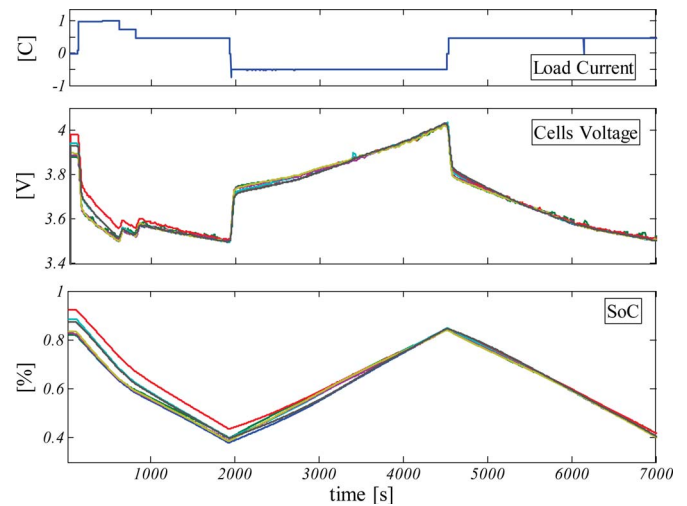


Fig. 9. Voltage-mode balancing. Positive current corresponds to the discharge phase, and negative current corresponds to the charge phase (each line represents one of the seven cells).

A. Balancing Procedure

The two decision strategies for identifying the cell to be bypassed have been evaluated. The test consists of randomly charging/discharging the battery pack for about 2 h and monitoring the balance state of each cell in terms of voltage and SoC. Fig. 9 shows the measured transients in the case of voltage analysis technique, and Fig. 10 shows those in the case of SoC analysis technique.

Whatever the balancing strategy is, it is evident that all cells remain balanced during the time regardless of the current sign and also in terms of the variable that is not directly controlled.

Both solutions are quite equivalent in terms of balancing effectiveness. Of course, the voltage approach has a slight structural advantage because it is much simpler to implement and it only requires voltage acquisition and no current sensing or SoC estimation, but it limits several possible control features. In fact, many different algorithms based on SoC estimation have been studied in order to model a battery pack [21] and, for example, predict the remaining battery capacity [22].

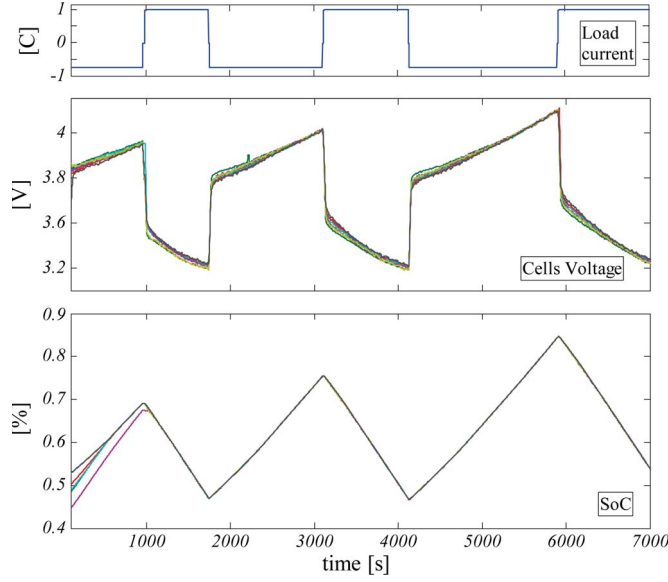


Fig. 10. SoC-mode balancing. Positive current corresponds to the discharge phase, and negative current corresponds to the charge phase (each line represents one of the seven cells).

B. Refresh Time Calculation

Data acquisition and SoC estimation are performed with time interval T_s , while pack configuration is refreshed with time interval T_{ref} .

Let us focus on the influence of T_{ref} on system performances. As the T_s value does not impact on the performance of the system as long as it remains small enough to allow an adequate acquisition rate, its value has been set constant to 2 s. On the contrary, T_{ref} is a critical parameter that hardly affects the performance of the system: Too large T_{ref} values correspond to a loss of balance accuracy, while too small T_{ref} values increase the stress of the system and cells (e.g., due to current spikes associated to the reconfiguration process) and electromagnetic interference emissions. An analytical estimation of the best value of T_{ref} has been investigated.

We consider a string of N perfectly balanced cells discharged at 1 C and a SoC-based balancing procedure. In order to compare performances obtained with different T_{ref} values, a figure of merit Q has been defined

$$Q := \frac{1}{N} \sum_{m=0}^{N-1} \frac{1}{N} \int_0^{(N-1)T_{\text{ref}}} |\text{SoC}_m(t) - \overline{\text{SoC}}| dt. \quad (3)$$

Parameter Q is the integrated absolute error in SoC, averaged across all the cells. This means that the balancing effect is better when the Q factor is low.

Fig. 11 graphically shows the terms of the evaluation of (3) for Q calculation in the case of $N = 7$ cells. Starting from the condition of all cells perfectly balanced with $\text{SoC} = \text{SoC}_i$ and considering the discharge phase of the pack, we consecutively disconnect one cell at a time for seven consecutive reconfigurations. Fig. 11 shows the superposition of SoC evolutions in time of a single cell $\text{SoC}_2(t)$ and the SoC mean value of the entire pack $\overline{\text{SoC}}$. The horizontal step represents the transient of the

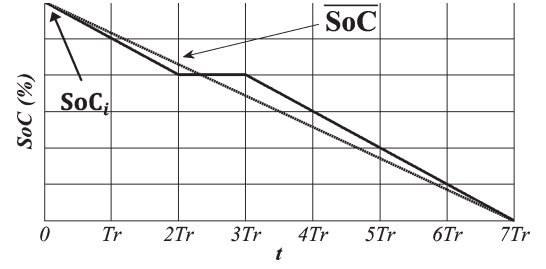


Fig. 11. Trend of the SoCs during a period of seven consecutive reconfigurations. T_r is the refresh time, and the vertical axis represents the SoC in arbitrary units.

disconnection of the considered cell. Numbering the cells from 0 to $N - 1$, the SoC of each m th cell depends on the time as

$$\text{SoC}_m(t) = \begin{cases} \text{SoC}_i - \alpha t, & t \in T1 \\ \text{SoC}_i - \alpha m T_{\text{ref}}, & t \in T2 \\ \text{SoC}_i - \alpha m T_{\text{ref}} - \alpha (t - (m+1)T_{\text{ref}}), & t \in T3 \end{cases} \quad (4)$$

where

$$\begin{cases} T1 = 0 \leq t \leq m T_{\text{ref}} \\ T2 = m T_{\text{ref}} \leq t \leq (m+1) T_{\text{ref}} \\ T3 = (m+1) T_{\text{ref}} \leq t \leq N T_{\text{ref}} \end{cases} \quad (5)$$

and α is a coefficient related to the discharge rate. Basically, it is the inverse of the time needed to fully discharge the battery at a given discharge rate. For example, if the battery is discharged at 1 C (the complete discharge is achieved in 1 h), $\alpha = 1/3600$ s.

The SoC mean value trend is a straight line described by the equation

$$\overline{\text{SoC}} = \text{SoC}_i - \alpha \left[\frac{N-1}{N} \right] t. \quad (6)$$

The deviation $\Delta \text{SoC}_m(t)$ of the SoC of the m th cell with respect to $\overline{\text{SoC}}$ results

$$\begin{aligned} \Delta \text{SoC}_m(t) &= \begin{cases} -\alpha \frac{1}{N} t, & t \in T1 \\ -\alpha \frac{1}{N} m T_{\text{ref}} + \alpha \frac{N-1}{N} (t - m T_{\text{ref}}), & t \in T2 \\ \alpha \frac{N-1-m}{N} T_{\text{ref}} - \alpha \frac{1}{N} (t - (m+1) T_{\text{ref}}), & t \in T3 \end{cases} \end{aligned} \quad (7)$$

and the mean value of $\Delta \text{SoC}_m(t)$ is

$$\overline{\Delta \text{SoC}_m} = \frac{1}{2} \frac{m^2}{N(N-1)} + \frac{1}{2} \frac{(N-1-m)^2}{N(N-1)}. \quad (8)$$

Finally, the value of the Q factor, i.e., the mean value over all cells, is

$$Q = \alpha \frac{1}{N} \frac{(2N-1)}{6} T_{\text{ref}}. \quad (9)$$

The Q parameter is directly proportional to the refresh time T_{ref} , i.e., an increase of the refresh time worsens the balancing effect. Moreover, increasing the number N of cells in the string worsens the balancing quality (even if not linearly). Fig. 12 shows the trend of the quality factor Q as a function

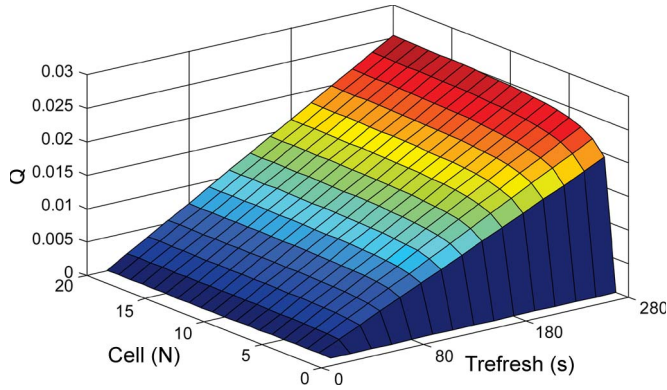


Fig. 12. Quality factor Q versus number of cells (N) and refresh time (T_{ref}). $\alpha = 1/3600$ s.

TABLE II
EXPERIMENTALLY MEASURED Q FACTOR (MEAN)
FOR DIFFERENT VALUES OF REFRESH TIME T_{ref}

T_{ref} (s)	Q
2	$2.6 \cdot 10^{-4}$
5	$1.4 \cdot 10^{-3}$
10	$1.5 \cdot 10^{-3}$
20	$2.2 \cdot 10^{-3}$
30	$2.8 \cdot 10^{-3}$
60	$4.2 \cdot 10^{-3}$
90	$7.8 \cdot 10^{-3}$
120	$9.5 \cdot 10^{-3}$

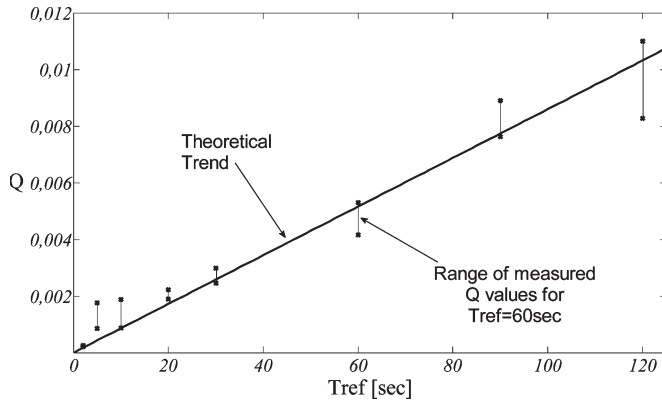


Fig. 13. Theoretical Q factor values versus experimental Q measurements. Sets of measured Q values at the same refresh time are represented with vertical bars that define their range of variation, i.e., the experimental uncertainty.

of the number of cells N and the refresh time T_{ref} (discharged at 1 C).

By using the realized prototype of the BMS, parameter Q has been experimentally obtained for different values of T_{ref} . Precisely, several values of Q have been measured with the same T_{ref} and different current profiles, and their mean value has been considered the final result. Measurements confirm the dependence of Q on T_{ref} , as summarized in Table II. The experimental results have been compared with the Q values expected by (9) for $N = 7$. Fig. 13 shows a comparison of the calculated and measured Q values. A good agreement is verified over all the T_{ref} considered ranges, both in terms of trend and variance of the measured data around the expected values.

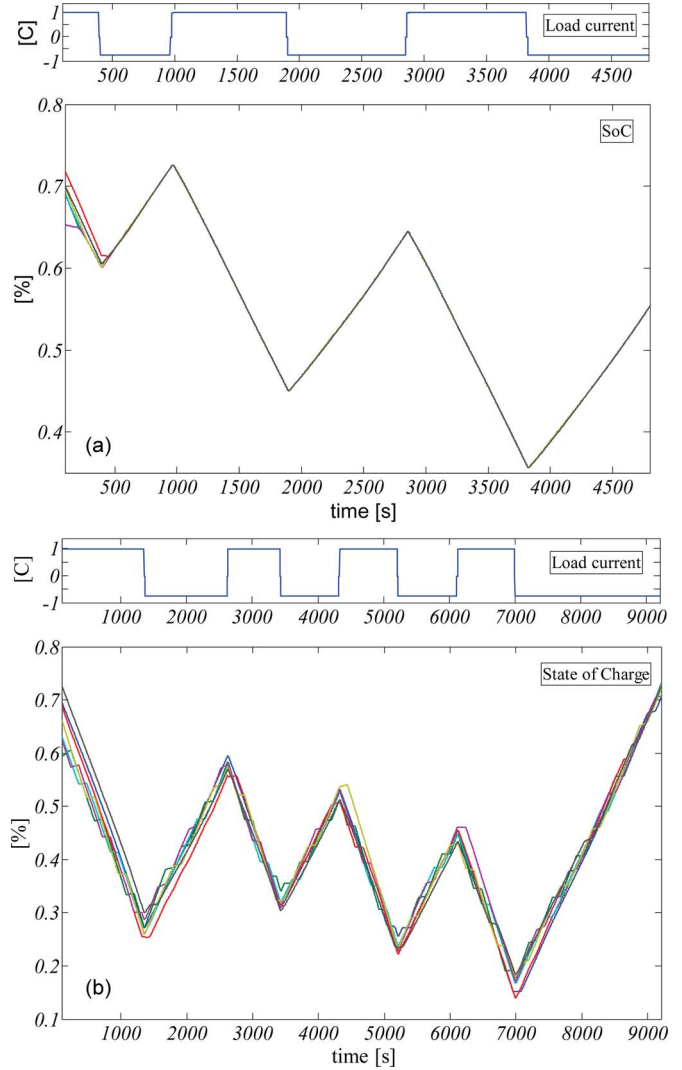


Fig. 14. Comparison of the SoC of a seven-cell string in the case of (a) $T_{\text{ref}} = 2$ s and (b) $T_{\text{ref}} = 120$ s. The plot shows also the considered load current profile (each line represents one of the seven cells).

In order to visually highlight the balance worsening as T_{ref} increases, Fig. 14 shows the comparison of the SoC of a seven-cell string in the case of $T_{\text{ref}} = 2$ s and $T_{\text{ref}} = 120$ s.

C. Efficiency

Efficiency is a key factor of any BMS architecture, and particularly for the presented solution. Even if, in principle, its calculation is straightforward due to the intrinsic simplicity of the layout, it has been a critical feature of the design process. In fact, switches that are connected in series to the current flow could determine a severe efficiency loss and overheating of the devices.

Fig. 15 shows the trend of efficiency versus voltage cell and battery pack current. The worst efficiency is at high current rates with low charged cells. On the contrary, the best performance is achieved with a low value of current and fully charged cells. It has been verified that the efficiency of the proposed BMS is always above 90% in any operative condition.

Although efficiency is a crucial parameter to compare a balancing system's performances, it is difficult to find numerical

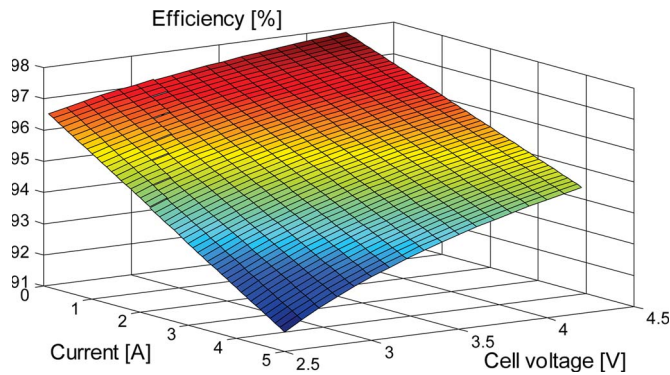


Fig. 15. BMS efficiency η versus cell voltage and load current.

values in the literature about it. This is because, depending on the balancing architecture, efficiency has a lot of dependences. Somewhere, it is dominated by dissipation on power components [23], and somewhere else, it is mainly due to microcontroller dissipation [24]. By the way, since most of the balancing architectures are based on a dc–dc converter, it can be supposed that their efficiency is comparable to that achieved by Ćuk or buck–boost systems (80%–95%). Using this realistic assumption, it can be stated that the presented solution provides a really high efficiency balance.

VI. CONCLUSION

A new BMS for string packs of Li-ion cells has been proposed. The core of the architecture is the technique of redundant cell, i.e., the reversible disconnection of a cell in the battery pack.

The main advantages of the proposed architecture are optimal balancing of the battery pack during operation and intrinsic hardness to cell faults. In fact, if a completely damaged cell is found in the pack, the system can permanently bypass it. Then, two scenarios can be selected. In one case, it is possible to continue the balancing operation since the system is designed to support multiple simultaneous disconnections. In this case, a voltage decrease on the overall pack voltage occurs. The second solution consists of stopping the balance action until the damaged cell is removed. In this case, no voltage reduction occurs.

Moreover, the proposed architecture is very flexible, economical, stable, and considerably space saving. In fact, compared with other balancing systems, it does not require any specific driver or PWM controller (that can strongly affect the overall cost), neither high-power inductors nor diodes (responsible for space consumption). The MOSFETs used to implement the switches do not need to meet any special requirement except the low ON-state resistance. Therefore, in the future, the development of a custom IC integrating the switch transistors (MOSFETs and BJTs) could be considered. A drawback is that a substantial effectiveness of the technique limits the dimension of the pack to about ten cells, i.e., up to 500 W. This means that the proposed BMS architecture is suited for low-power automotive applications, such as electric scooters and bikes. For higher power demanding loads, a modular combination of several cell-redundant BMSs can be conceived. Of course, in this case, the complexity of the layout is also sensibly increased.

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Antonio Manenti was born in Bergamo, Italy, in 1983. He received the M.Sc. degree in electrical engineering from Politecnico di Milano, Milano, Italy, in 2007, where he has been working toward the Ph.D. degree in the Dipartimento di Elettronica e Informazione since 2008.

He was a Visiting Scholar with the Center for Automotive Research, The Ohio State University, Columbus, from May 2010 until December 2010. His research is focused on power electronics for electric and hybrid vehicles, as well as battery model

design.



Andrea Abba was born in Lecco, Italy, on May 24, 1983. He received the M.Sc. degree from Politecnico di Milano, Milano, Italy, in 2007, where he has been working toward the Ph.D. degree in the Dipartimento di Elettronica e Informazione since 2008.

His research is focused on digital nuclear spectroscopy and advanced digital systems for audio and video processing. He is the author of 18 international publications.



Alessandro Merati was born in Bergamo, Italy, in 1985. He received the B.Sc. degree in computer science from Politecnico di Milano, Milano, Italy, in 2008, where he is currently working toward the M.Sc. degree in electronics engineering, working on the design and characterization of battery management systems for electric vehicles.



Sergio M. Savaresi (M'00) was born in Manerbio, Italy, in 1968. He received the M.Sc. degree in electrical engineering and the Ph.D. degree in systems and control engineering from Politecnico di Milano, Milano, Italy, in 1992 and 1996, respectively, and the M.Sc. degree in applied mathematics from Catholic University, Brescia, Italy, in 2000.

After the Ph.D. degree was received, he worked as a Management Consultant at McKinsey & Company, Milano. Since 2006, he has been a Full Professor of automatic control with Politecnico di Milano, where he is also the Head of the "Motor Vehicle" research team (<http://move.dei.polimi.it>). He is the author of more than 250 scientific publications at international level (among which are many patents), and he has been the Proposer and Manager of more than 50 sponsored joint research projects between Politecnico di Milano and private companies. His main interests are in the areas of vehicle control, automotive systems, data analysis and system identification, nonlinear control theory, and control applications.

Dr. Savaresi is an Associate Editor of the IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY, the *European Journal of Control*, the *IET Control Theory and Applications*, and the *International Journal of Vehicle Systems Modelling and Testing*. He is also a member of the Editorial Board of the IEEE Control Systems Society.



Angelo Geraci (M'01–SM'03) was born in Milano, Italy, in 1967. He received the Laurea degree (*cum laude*) in electronic engineering and the Ph.D. degree (*cum laude*) in electronics and communication engineering from Politecnico di Milano, Milano, in 1993 and 1996, respectively.

He has been a Scientific Collaborator with the Italian National Institute of Nuclear Physics (INFN), Frascati, Italy, since 1995. He has been an Assistant Professor since March 1999 and an Associate Professor since December 2004 with the Dipartimento di Elettronica e Informazione, Politecnico di Milano. He is the author of more than 100 publications in international congress proceedings and journals.

Dr. Geraci has been a Senior Member of the IEEE Nuclear and Plasma Sciences Society since February 2003. He was the recipient of the 2004 Prize Paper Award from the IEEE Power Electronic Society.